

Accident prevention through visual warnings

How to design warnings in head-up display for older and younger drivers

Juela Kazazi, Susann Winkler & Mark Vollrath

Technische Universität Braunschweig

Institut für Psychologie

Abteilung für Ingenieur- und Verkehrspsychologie

Braunschweig, Germany

Abstract— Many severe accidents occur in urban areas. As part of the research project UR:BAN this study investigated how different types of visual warnings can prevent collisions within these areas and trigger an adequate driver reaction. For this, two different warning types were implemented in a head-up display. It was assumed that one of the warning types would trigger an immediate and strong brake reaction of the driver (stop sign warning). The second warning type should animate the driver to be cautious, for example by slightly pressing the brake pedal (caution sign warning). As collisions in urban areas are very severe for older drivers, differences in the driving performances (numbers of collisions, brake reaction time, maximum braking value) of older and younger drivers due to the warning types were investigated. For this, four urban scenarios were implemented in a fixed base driving simulator varying the characteristics of the critical object (e.g., pedestrian, lead vehicle, obstacle). In total 72 drivers (36 participants aged 20-35 years, 36 participants aged 65+ years) were tested in a between-subjects design (type of warning, age). The study revealed that the number of collisions was reduced when drivers were warned (especially younger drivers) demonstrating the positive effect of warnings in very critical scenarios. In less critical situations, the warnings did not reduce the numbers of collisions, since the amount of collisions was already low. In critical scenarios older drivers preferred the stop sign warning (leading to a faster and stronger brake reaction) compared to younger participants. Younger drivers had the fastest and strongest brake reaction when receiving the caution sign warning. In less critical scenarios both warnings had a positive effect on the braking reaction time for younger and older drivers. In both groups the strongest brake reaction was reached when receiving the stop sign warning. This study demonstrates the importance of warning design when wanting to create effective warnings.

Keywords—Driver Warning, Accident Prevention, Head-up Display, Older and Younger Drivers; Simulator Study; Urban Area

I. INTRODUCTION

For most drivers the urban area is a very challenging one resulting in a large number of accidents. Within these areas the load of information is very high and the attention has to be divided, since many traffic elements (oncoming vehicles, traffic lights) need to be perceived simultaneously. Currently assistance systems are progressively being implemented in vehicles in order to make urban areas safer. Yet, when developing assistance systems some imperative factors need to be considered [2]. Some of these factors include: location

(head-up display (HUD), dashboard), modality (visual, auditory, haptic), type of information (text, symbols).

The display location of assistance systems is important to support and not distract drivers [8], [1], [11]. Information or warning of assistance systems are located in different areas within the vehicle, such as, for example, in head-down displays (HDD) as well as head-up displays (HUD). Assistance systems in HDD force drivers to take their eyes off the road in order to get the provided information. Accident occurrence increases with the time the eyes are off of the road [14], [12], and [21]. Reference [14] conducted a simulator study with truck drivers, who were asked to keep their speed close to either 64 km/h or 112 km/h. Drivers controlled their speed better with information displayed on a HUD rather than on a HDD. The study of [15] has shown that with a HUD the glance behavior of participants was relatively short, since drivers are not forced into taking their eyes off of the road. It could also be shown that drivers had a gain in reaction time by 0.8 to 1.0 seconds with a HUD compared to a HDD. Furthermore, the HUD was rated as being “extremely meaningful” (close to 5, on a 1 to 5 scale). These studies have revealed some of benefits of HUD compared to HDD.

Many studies focus on how to best assist the driver. Yet, research on how to warn the driver and how warnings should be designed in the HUD are rare. Some researchers consider traffic signs as a type of warning to be intuitive. Drivers are confronted with these signs every day and should have an association of reaction with them [17] and [16]. Since there are many different traffic signs, the question arises whether some traffic signs are better suited than others when warning drivers and whether different reactions are triggered by different traffic signs. It is hypothesized that a stop sign will evoke a fast and strong braking, since this sign is linked to an immediate reaction. A caution sign, on the other hand, is hypothesized to cause the driver to be rather attentive without a harsh braking reaction.

When developing warning systems the age of drivers is not to be neglected since many studies have shown that the driving behavior as well as the information processing rate of older and younger drivers differs. Reference [17] reported that older drivers have a reduced information processing rate. They performed worse in all laboratory information processing test compared to younger drivers. Furthermore it was reported that older drivers had a reduction in the speed of performance (slower perception and processing of viewed information), as

well as the possibility of overstrain in complex and critical situations [18]. Concerning the driving behavior, for example [23] examined the effects of traffic speed, driver age and gender on left turn gap acceptance behavior, drivers' acceleration rate and steering action at a stop-controlled intersection in a driving simulator experiment. A total of 63 participants (young: 20-30 years; mid aged: 31-55 years; old: 56-83 years) had to turn left with slow moving traffic (40.2 km/h) as well as fast moving traffic (88.5 km/h). The results showed significant age effects (e.g. older drivers accelerated at a slower pace). Yet, older drivers may compensate for their cognitive impairments by simplifying their driving task (e.g. driving slower as compared to younger drivers) [4], [17], [9], [20], [6] and [23]. From this, the question rises, if the symbols of warning systems need to be different for older and younger drivers.

As part of the research project UR:BAN the following study investigated how warnings should be designed in urban areas to warn different drivers of two different age groups (older drivers vs. younger drivers).

The focus of this research was to determine:

- if warnings would help reduce older and younger drivers' crashes in different critical urban scenarios,
- if warning symbols need to be different for older and younger drivers,
- what kind of warning symbols might have the best effect (e.g., brake reaction time and maximum braking value)

For this, two simulator studies were conducted (older drivers vs. younger drivers). These studies will be explained in more detail in the next section.

II. METHOD

A. Participants

Two groups of drivers (older and younger drivers) were examined. The older participants group contained 36 drivers (29 male, 7 female) all being above 65 years old and having a mean age of 71.9 years (SD = 4.4 years) as well as an annual mileage of about 12,000 km. This study was conducted in the time of March 2014 to May 2014. All 36 younger participants (20 male, 16 female) were above 18 years old, had a mean age of 23.9 years (SD = 4.2 years) and drove about 6000 km annually. This study was conducted from July 2013 to October 2013. Participants were recruited from a database of the Technische Universität Braunschweig, consisting of younger and older drivers. All drivers were trained in the driving simulator in order to ensure that they were familiar and felt comfortable driving in the simulator. This training (containing three parts, all being the same for older and younger drivers) was conducted on the same day as the experiment to increase compliance. In the first and second part of the training, the participants drove on a straight rural road followed by an urban road in order to train acceleration and deceleration. The last part contained urban intersections where the participants had to turn left/right, in order to get familiar with the turning behavior. Participants who developed simulator sickness had to

be excluded from the study (16 older and 2 younger drivers). All 72 remaining participants had normal or corrected-to-normal vision. They were compensated for their time by either receiving a box of chocolates, eight Euros an hour or course credit (if university students).

B. Driver Warnings

Two different warning signs were considered in this paper. Drivers were warned either with a stop sign or a caution sign in the HUD. The HUD was located above the roadway, where it would not obscure the view of the driver (traffic lights or signs). Many warning systems are coupled with an auditory signal [10] but a high frequency of auditory signals may become annoying for drivers [22]. Furthermore the focus of this study lay on the effect of visual warnings, this being the reasons that auditory signals are not considered. The duration of the warning was individual and depended on the speed of the drivers. The onset as well as the offset of the warning was triggered at particular flow points. If a participant came to a complete stop between the two flow points, the duration of the warning was longer as compared to drivers, who did not come to a stop between these points. In total, there were four experimental groups. One group comprised of older drivers only receiving the stop sign warning (SW) in the HUD. The other group of older drivers received the caution sign warning (CW). Younger participants were also divided into two groups, receiving either the SW or the CW warning. Each age group had their own control group (C) receiving no warning at all (see Table I). The size of the SW warning in the HUD was 19.9x17.7 cm; the CW warning was 21x19 cm.

It was assumed that the SW warning would provoke an immediate brake reaction of drivers, whereas the CW would guide the attention of the driver onto the critical object. It was expected that the different symbols (pedestrian, vehicle or obstacle) would help drivers to identify the critical object and react on it by being cautious and not showing an immediate strong reaction.

WARNING TYPES AND WARNING SYMBOLS

Type of Warning	Presentation in head-up display		
Control	No warning sign		
Stop sign warning (SW)			
Caution sign warning (CW)			

C. Driving Scenarios and Simulation

The different scenarios, as seen in Table II, were created using the driving simulation software SILAB [13]. Table I gives an overview of all scenarios and their criticality, as well as the timing of the warnings. The timing of the warnings was the same for older and younger drivers in each scenario, except the *Pedestrian 2* scenario. The results of the previous study only with younger participants (which was conducted formerly) showed that the timing of the warning was very late. Thus, the timing of the warning was changed for the following

study with older drivers (warning was now triggered seven meters earlier).

In each scenario the ego vehicle, driven by the participant, traveled through a simulated urban road and had to make a turn when it was indicated by a voice output and an arrow in the speedometer. Scenarios were implemented to be different in their criticality, in order to find out what type of warning was best suited for very critical to less critical situations. In the *Pedestrian 1* scenario, the driver had to make a left turn at an intersection. While turning, a pedestrian crossed the ego vehicles road. This scenario was implemented to reconstruct a critical traffic situation. In the *Pedestrian 2* scenario, a pedestrian hidden by a parked car suddenly crossed, making this scenario a very critical situation. The *Vehicle* scenario was a less critical situation where a lead vehicle came to a sudden stop. In the less critical *Obstacle* scenario, drivers were confronted with an obstacle (hay bale) hidden behind a hill.

TABLE I. CRITICALITY OF SCENARIOS AND TIMING OF WARNING USED IN THE STUDY

Name	Criticality	Picture	Time of triggered warning
Pedestrian 1	Critical		Start of warning three meters after the Pedestrian was activated
Pedestrian 2	Very critical		For older drivers the warning was triggered seven meters before the pedestrian was activated. For younger drivers the warning and the moving of the pedestrian were activated at the same time
Vehicle	Less critical		Start of warning seven meters before the lead vehicle started to brake, independantly of the gap between the ego vehicle and the lead vehicle
Obstacle	Less critical		Warning is triggered 70 meters before the hay bale becomes visible

The study was conducted in the fixed base medium fidelity driving simulator of the Technische Universität Braunschweig. The scenery is projected by three beamers on three screens (front, left, right) resulting in a 180 degree field of view. During this study an automatic transmission was used. Rear-view mirrors are simulated by three LCD screens (front, left, right). By using a driving simulator an accurate measure of driving performance is possible and essential variables (road type, weather, day time) can be controlled. Furthermore an evaluation of warning systems in critical situations is made possible without harming the driver.

D. Procedure

After reading and signing a consent form, participants were instructed in written form about the procedure of the experiment. Drivers were told to “drive as they would normally do in their own vehicles” and to obey all traffic rules (e.g., speed limits). Afterwards, drivers completed the training drive (lasting about 25 min) in order to get familiar with the simulator and prevent simulator sickness. During the whole experiment the investigator was seated in a separate room, having the opportunity to communicate with the participant via a microphone. The test drive lasted about 15 min. After that the subjects were debriefed about the purpose of the study, reimbursed and thanked for their participation. The overall duration of the trial was about two hours. The order of

scenarios in the test drive was not counterbalanced, since the effects of the warning types were of interest.

E. Data and Data Analysis

Driver behavior was recorded by the simulation software SILAB [13]. Table III gives a summary of the dependent variables.

For data analysis IBM SPSS Statistics 22 was used. When applicable, a general linear model, with two between-subject factors (*Type of Warning* and *Age*) was used. For accident occurrence, chi-squared tests were conducted. A significance level of $\alpha p = .05$ was adopted for all statistical tests.

TABLE II. DEPENDENT VARIABLES REGARDED IN THIS STUDY

Variable	Unit	Description
Number of collisions	-	Number of collisions that occurred between the ego vehicle and the critical object
Brake reaction time	s	Reaction time to pressing the brake pedal after the warning was triggered
Maximal braking value	%	Maximal value of brake pedal pressing

III. RESULTS

A. Number of collisions

Older drivers had a low number of collisions, regardless of the presence and type of warning, when compared to younger drivers (see Fig. 1).

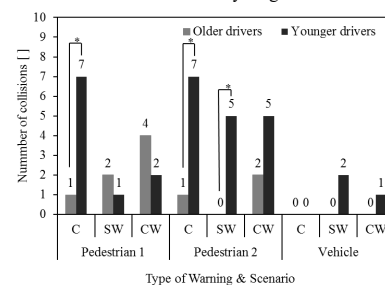
There were no significant differences in the number of collisions between older and younger drivers in the SW and CW group (SW: $\chi^2 = 0.38, p = 1.000$; CW: $\chi^2 = 0.89, p = .640$) in the *Pedestrian 1* scenario. However, younger drivers in the control group had significantly more collisions compared to older drivers ($\chi^2 = 6.75, p = .027$).

In the *Pedestrian 2* scenario the amount of collisions for younger drivers was relatively high in the experimental groups, this being the reason for the changed timing of the warning (see section: *Driving Scenarios and Simulation*). Younger drivers in the control group had significantly more collisions compared to older drivers in the control group ($\chi^2 = 6.75, p = .027$). Furthermore, younger drivers had significantly more accidents in the SW compared to older ones ($\chi^2 = 6.32, p = .037$).

In the *Vehicle* scenario, there were no significant differences between younger and older drivers (SW: $\chi^2 = 2.18, p = .478$; CW: $\chi^2 = 1.04, p = 1.000$).

In the *Obstacle* scenario there were no collisions at all. Fig. 1 gives an overview of the numbers of collisions in *Pedestrian 1*, *Pedestrian 2* and *Vehicle* scenario, where the * stands for $p > .05$.

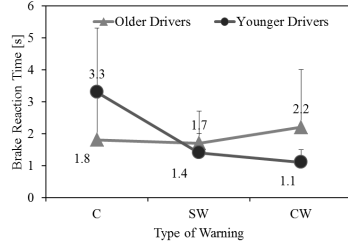
Fig. 1: Number of collisions in the *Pedestrian 1*, *Pedestrian 2* and *Vehicle* scenario for older and younger drivers



B. Brake Reaction Time

For this analysis a general linear model, with two between-subject factors (*Type of Warning* and *Age*) was used. For this analysis not all participants were considered. If participants had not pressed the brake pedal and did not have a collision, they were excluded from the examination, as the scenario had not functioned as planned. If participants had not pressed the brake pedal but had a collision, their brake reaction was replaced with the longest brake reaction of their individual group as this failure to react is similar to a very long reaction time.

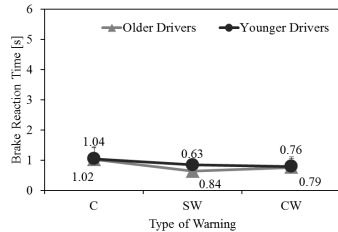
Fig. 2: Brake Reaction Time in the *Pedestrian 1* scenario for older and younger drivers



In the *Pedestrian 1* scenario there was an interaction between the factors *Type of Warning* * *Age*. Older drivers had the fastest brake reaction time with SW compared to younger drivers who reached their fastest brake reaction time when the CW was presented (see Fig. 2). Moreover, for younger drivers both warnings were effective in reducing brake reaction time while this was not the case for older drivers who had even longer brake reaction times with CW than in the control group.

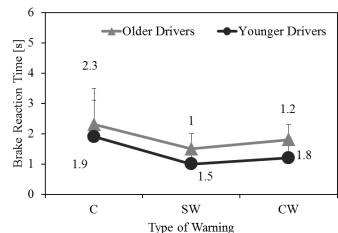
During the *Pedestrian 2* scenario there was a main effect of the factor *Type of Warning*, as seen in Fig. 3. Here the control group had significantly the slowest brake reaction time compared to the SW ($p = .001$) as well as compared to the CW ($p = .003$) group.

Fig. 3: Brake Reaction Time in the *Pedestrian 2* scenario for older and younger drivers



In the *Vehicle* scenario a main effect of the factor *Type of Warning* as well as *Age* was found, as seen Fig. 4. The control group had the slowest brake reaction time compared to the SW ($p < .001$) and the CW ($p = .004$) group. Older drivers had an overall faster braking reaction compared to younger drivers

Fig. 4: Brake Reaction Time in the *Vehicle* scenario for older and younger drivers



During the *Obstacle* scenario there was only a main effect of the factor *Type of Warning*. Here again the control group had the slowest brake reaction time compared to the SW ($p < .001$) and the CW ($p = .009$) group. There was a tendency to a significant difference between the SW and CW group, with the SW having faster brake reactions ($p = .055$). The age groups did not differ.

Fig. 5: Brake Reaction Time in the *Obstacle* scenario for older and younger drivers

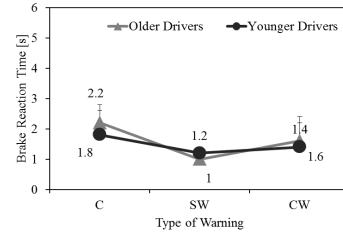


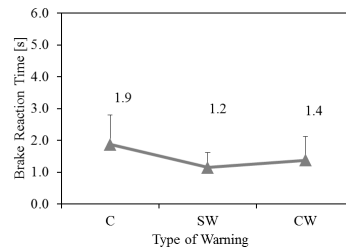
Table 5 gives an overview of the test results of the brake reaction times in all four scenarios.

TABLE III. RESULTS OF BRAKE REACTION TIME IN ALL FOUR SCENARIOS

Brake Reaction Time				
Between-subject: Type of Warning (W)				
	F	df	p	η^2
Pedestrian 1	3.68	2,60	.032	.120
Pedestrian 2	7.59	2,72	.001	.187
Vehicle	9.71	2,62	.000	.258
Obstacle	10.85	2,70	.000	.253
Between-subject: Age (A)				
	F	df	p	η^2
Pedestrian 1	0.00	1,60	.987	.000
Pedestrian 2	1.76	1,72	.189	.026
Vehicle	10.66	1,62	.002	.160
Obstacle	0.42	1,70	.521	.006
W * A				
	F	df	p	η^2
Pedestrian 1	4.64	2,60	.014	.147
Pedestrian 2	0.84	2,72	.435	.025
Vehicle	0.09	2,62	.914	.003
Obstacle	0.82	2,70	.447	.025

For an overall description of the effect of the warning, we computed the mean of the braking reaction time for each subject over all scenarios. Fig. 6 gives the results. The brake reaction time is reduced from 1.9 s in the control group to 1.2 s in the SW group and 1.4 s in the CW group. Thus, overall there is a slight advantage of the SW type of warning.

Fig. 6: Mean Brake Reaction Times over all scenarios and all drivers in each Type of Warning

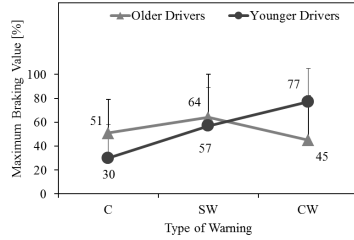


C. Maximum Braking Value

For this analysis a general linear model with two between-subject factors (*Type of Warning* and *Age*) was used. The maximum braking value gives information about the amount (in %) of pressing the pedal. This value demonstrates if the driver either pressed the brake pedal slightly or firmly. The higher the percentage of the maximum braking value, the firmer the brake pedal was pressed. For this analysis not all participants were considered. If participants did not press the brake pedal and did not have an accident, they were excluded from the examination, leading to dissimilar numbers of participants in the groups. If participants did not press the brake pedal but had a collision, as their maximum braking value the smallest maximum braking value of the individual group was used.

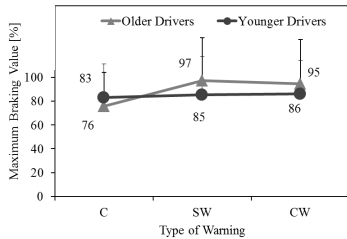
In the *Pedestrian 1* scenario an interaction between *Type of Warning* * *Age* was found. Older drivers reached their highest maximum braking value in the SW, while younger participants had their highest value in the CW, as seen in Fig. 7. For older drivers the CW caused a lesser maximal braking values compared to the control group. Overall, only the SW warning produced larger values than the control group.

Fig. 7: Maximum Braking Value in the *Pedestrian 1* scenario for older and younger drivers



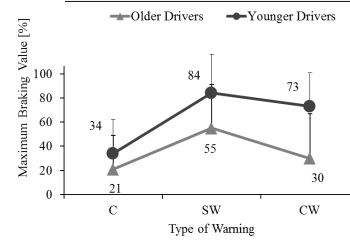
In the *Pedestrian 2* scenario a main effect of *Type of Warning* was found (see Fig. 8). The control group had lesser maxima compared to the SW ($p = .031$) and the CW ($p = .046$) group in both age groups.

Fig. 8: Maximum Braking Value in the *Pedestrian 2* scenario for older and younger drivers



During the *Vehicle* scenario a main effect of both the factors *Type of Warning* and *Age* was found (Fig. 9). Participants in the SW ($p < .001$) and CW ($p = .019$) group had higher maxima's compared to the control group. Furthermore older drivers had overall smaller maxima compared to younger drivers (older drivers: $M = 35.6\%$, $SD = 37.6$ vs. younger drivers: $M = 65.2\%$, $SD = 36.4$).

Fig. 9: Maximum Braking Value in the *Vehicle* scenario for older and younger drivers



In the *Obstacle* scenario the factor *Type of Warning* showed a main effect. Compared to the SW group, the control group had significantly lower maxima's ($p < .001$). No significant difference could be found between the control and the CW group (Fig. 10). Yet, the SW group had significantly higher maxima compared to the CW group ($p < .001$).

Fig. 10: Maximum Braking Value in the *Obstacle* scenario older and younger drivers

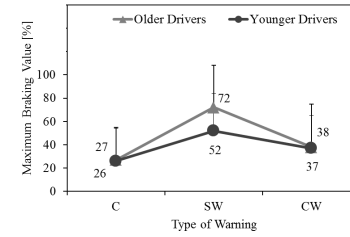


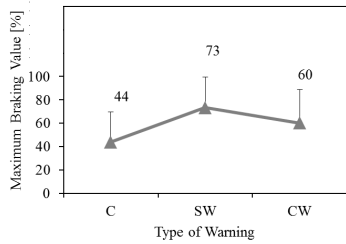
Table 5 gives an overview of the test results of the maximum braking value in all four scenarios.

TABLE IV. RESULTS OF MAXIMUM BRAKING VALUE IN ALL FOUR SCENARIOS

	Maximum Braking Value			
	Between-subject: Type of Warning (W)			
	F	df	p	η^2
Pedestrian 1	2.85	2,64	.066	.089
Pedestrian 2	3.02	2,72	.056	.084
Vehicle	8.70	2,64	.000	.231
Obstacle	17.14	2,70	.000	.349
	Between-subject: Age (A)			
	F	df	p	η^2
Pedestrian 1	0.20	1,64	.888	.000
Pedestrian 2	1.10	1,72	.298	.016
Vehicle	11.91	1,64	.001	.170
Obstacle	1.95	2,70	.167	.030
	W * A			
	F	df	p	η^2
Pedestrian 1	3.89	2,64	.026	.118
Pedestrian 2	1.86	2,72	.164	.053
Vehicle	1.05	2,64	.356	.035
Obstacle	1.70	2,70	.190	.051

Overall, both warnings were somewhat effective as compared to the control group in increasing the braking reaction of the drivers. However, this effect was stronger for the SW warning as compared to the CW warning. This was found for both age groups. Additionally, there were some differences between the two age groups with regard to the amount of braking in some scenarios.

Fig. 11: Mean Maximum braking value over all scenarios and all drivers in each Type of Warning



IV. DISCUSSION

The study examines the effect of two types of warning in different critical scenarios for older and younger drivers. Before discussing the results, some caveats have to be considered. Comparing the data of different participants is only valid to a limited degree due to the different timings of the warning. The timing of the warning was not independent of the drivers vehicle speed. For further studies the time to collision (TTC) should be used as the time to trigger a warning, as participants would then receive the warning at the same “critical” level. Furthermore, comparing the results of older and younger drivers in the *Pedestrian 2* scenario is difficult, as the time of the triggered warning was different for older and younger drivers. Yet, when looking at the control group in this scenario, it becomes clear that younger drivers have significantly more collisions compared to older drivers independently of the timing of the warning. For further studies the timing of the warning should be the same for all groups.

During this study older drivers had less accidents compared to younger drivers. One possibility might be that older drivers had a higher annual mileage and thus more driving experience compared to younger drivers. Other reasons for the low accident occurrences of older drivers might have been compensatory strategies like driving slower. Additionally, a selection bias might have resulted and those older drivers volunteering for this study were all in good shape. For further studies more female older drivers should be recruited, since the amount of female older drivers in this study was relatively low.

The four implemented scenarios were designed to vary in their criticality. This could be shown when looking at the maximal braking values in each scenario (Older Drivers: *Pedestrian 1*: 85%; *Pedestrian 2*: 54%; *Vehicle*: 36%; *Obstacle*: 45% vs. Younger Drivers: *Pedestrian 1*: 89%; *Pedestrian 2*: 55%, *Vehicle*: 65%; *Obstacle*: 39%). The reasons for the different criticalities of older and younger drivers in the *Vehicle* scenario are that older drivers had a significant lower speed when the warning was triggered compared to younger drivers ($F_{(1, 71)} = 22.6, p < .001$; older drivers: $M = 11.3, SD = 1.9$ vs. younger drivers: $M = 12.9, SD = 0.7$). Furthermore older drivers had a significant larger distance to the lead vehicle when pressing the brake pedal ($F_{(1, 61)} = 16.51, p < .001$; Older Drivers: $M = 53.3, SD = 30.0$ vs. Younger Drivers: $M = 28.2, SD = 17.8$). Additionally the different criticalities of the scenarios become clear when looking at the number of collisions. The most accidents occurred in the *Pedestrian 2*

(critical) and *Pedestrian 1* (very critical) scenario. Furthermore, it could be shown that the SW lead to the fastest and strongest brake reactions. Yet, the question arises, whether drivers need to have a very high braking maximum even if the situation is not critical and a sudden and firm brake reaction is not required, as in the *Vehicle* and *Obstacle* scenario. Here the SW could lead to negative effects (e.g., maximum braking value is too high and not suited for a less critical situation) so that other road users might perceive this behavior as traffic blocking, leading to hazardous behaviors of other drivers or resulting in rear-end crashes [5]. In order to avoid that, a warning cascade should be considered with different warnings depending on the criticality of the situation as well as the reaction required. This will be discussed in the next section.

V. CONCLUSION

The aim of this work was to determine if warning systems would help reduce older and younger driver’s crashes in urban scenarios. Furthermore the focus laid on the warning symbols as well as what kind of warning symbols might have the best effect on brake reaction time and maximum braking value in critical urban situations.

The reduced number of accidents, especially for younger drivers, demonstrates the positive effects of these warning systems in (very) critical scenarios. When the situation is not critical, then warning systems did not have a positive effect on the reduction of collisions, since the amount of collisions was already low.

In the very critical scenario *Pedestrian 2*, both warning types lead to faster reaction times, as well as maximal braking values compared to the control group, independently of the age of drivers.

In the less critical *Vehicle* scenario both warning types had a positive effect on the brake reaction time for older drivers. Furthermore older drivers reacted faster compared to younger drivers independently of the warning type, at least in this scenario. Yet, they had a lower maximum braking value, since their velocity was significantly lower compared to younger drivers. The highest maximum braking value was reached (in both groups) when receiving the SW.

In the less critical *Obstacle* scenario, the SW lead to the fastest brake reaction as well as to the highest maximum braking value in both groups.

In summary, the warnings used in this study had a positive effect on the reduction of accidents for younger drivers. In critical situations (e.g., *Pedestrian 1*) older drivers should receive the SW in order to react fast and firm. Younger drivers should receive the CW in these critical situations. For very critical scenarios (e.g., *Pedestrian 2*) older and younger drivers should receive any kind of warning (no difference between SW and CW). In less critical scenarios (*Vehicle* or *Obstacle*) the CW is best suited for both groups, since a high maximum braking value is not required.

The descriptive results of the warnings, independently of the scenario and the age of drivers, showed that the SW lead to the fastest reaction time as well as the highest maximum

braking values. This type of warning can be recommended for quite critical scenarios, which demand an immediate driver reaction. In less critical situation the CW should be triggered, since raising the attention of the driver and a slight braking reaction are sufficient reactions. When drivers do not react in less critical situations, a warning cascade combining both warning types would be reasonable [3], since the situation might become dangerous. If the driver reacts upon the CW as a first prewarning stage (e.g. slight braking) the SW will not follow. Yet, if the driver did not react upon the CW and the situation is about to get critical, the SW warning is to be triggered. This warning cascade will be examined in further studies, focusing on whether it is suitable for older and younger drivers.

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